

AI Response Quality Evaluator

Batch Evaluation Report

This report presents an automated quality assessment of AI-generated responses using a multi-agent evaluation framework. Each response is independently analyzed across four quality dimensions before producing an overall verdict.

Generated On	06 August 2026, 10:30 PM
Report Type	Batch Evaluation
Evaluation Engine	Multi-Agent LLM Evaluator
Framework	Retrieval-Augmented Generation (RAG)
LLM Provider	Groq

Total Responses	Average Score
10	7.2/10
Pass	Needs Improvement
1	7

Evaluation Dimensions

- ✓ Relevance
- ✓ Accuracy
- ✓ Hallucination Detection
- ✓ Completeness

This report was generated automatically by the AI Response Quality Evaluator.

Executive Summary

This report presents the overall performance of the AI Response Validation System with Hallucination Detection Assistance. The submitted responses were evaluated using multiple specialized agents to measure relevance, accuracy, hallucination detection and completeness before generating an overall quality verdict.

Batch Statistics

Metric	Value
Total Responses	10
Successfully Evaluated	10
Average Score	7.20/10
Pass	1
Needs Improvement	7
Fail	2

Dimension-wise Performance

Dimension	Average Score
Relevance	7.7/10
Accuracy	6.7/10
Hallucination	7.3/10
Completeness	7.4/10

Key Findings

- Average Overall Score: **7.20/10**
- Best Performing Dimension: **Relevance**
- Weakest Dimension: **Accuracy**
- Hallucination Frequency: **90.0%**
- Highest Response Score: **9.6/10**
- Lowest Response Score: **2.2/10**

Overall Assessment

The evaluated responses demonstrate satisfactory overall quality. While most responses are relevant and factually reasonable, improvements are recommended in weaker evaluation dimensions to further increase response reliability.

Batch Analytics

Metric	Value
Average Relevance	7.7/10
Average Accuracy	6.7/10
Average Hallucination	7.3/10
Average Completeness	7.4/10
Highest Overall Score	9.6/10
Lowest Overall Score	2.2/10
Best Performing Dimension	Relevance
Weakest Dimension	Accuracy
Hallucination Frequency	90.0%
Grounded Responses	1

Verdict Distribution

Verdict	Count	Percentage
Pass	1	10.0%
Needs Improvement	7	70.0%
Fail	2	20.0%

Summary

A total of **10** responses were processed. **10** responses completed successfully. The average overall evaluation score was **7.2/10**. The strongest evaluation dimension was **Relevance**, while the weakest was **Accuracy**. The detected hallucination frequency across the successfully evaluated responses was **90.0%**.

Individual Evaluation Results

Response #1

Question

What is Retrieval-Augmented Generation?

AI Response

RAG combines information retrieval with generative language models.

Dimension	Score
Relevance	9/10
Accuracy	10/10
Hallucination	10/10
Completeness	9/10
Overall	9.6/10

Pass

Response #2

Question

What is Machine Learning?

AI Response

Machine learning is a branch of AI where systems learn patterns from data instead of relying only on explicitly programmed rules.

Reference Answer

Machine Learning enables computers to learn from data and improve performance without explicit programming.

Dimension	Score
Relevance	9/10
Accuracy	8/10
Hallucination	7/10
Completeness	9/10
Overall	8.2/10

Needs Improvement

Unsupported Claims

- Machine learning is a branch of AI where systems learn patterns from data instead of relying only on explicitly programmed rules.

Response #3

Question

What is overfitting in machine learning?

AI Response

Overfitting happens when a model memorizes the training data, including noise, and performs poorly on unseen data.

Reference Answer

Overfitting occurs when a model learns training data too closely and fails to generalize.

Dimension	Score
Relevance	9/10
Accuracy	6/10
Hallucination	7/10
Completeness	9/10
Overall	7.6/10

Needs Improvement

Unsupported Claims

- Overfitting happens when a model memorizes the training data, including noise, and performs poorly on unseen data.

Response #4

Question

What is the purpose of normalization?

AI Response

Normalization scales numerical features to a common range, helping many algorithms converge faster and perform better.

Reference Answer

Normalization rescales feature values to improve training stability and model performance.

Dimension	Score
Relevance	9/10
Accuracy	8/10
Hallucination	7/10
Completeness	9/10
Overall	8.2/10

Needs Improvement

Unsupported Claims

- Normalization scales numerical features to a common range, helping many algorithms converge faster and perform better.

Response #5

Question

What is a database index?

AI Response

A database index is like a table of contents that speeds up data retrieval, although it may slightly increase storage and update costs.

Reference Answer

Indexes improve query performance by organizing references to data.

Dimension	Score
Relevance	9/10
Accuracy	6/10
Hallucination	7/10
Completeness	9/10
Overall	7.6/10

Needs Improvement

Unsupported Claims

- A database index is like a table of contents that speeds up data retrieval, although it may slightly increase storage and update costs.

Response #6

Question

Explain cloud computing.

AI Response

Cloud computing means storing files on Google Drive.

Reference Answer

Cloud computing provides on-demand computing resources such as servers, storage, databases, and networking over the internet.

Dimension	Score
Relevance	4/10
Accuracy	2/10
Hallucination	7/10
Completeness	1/10
Overall	3.4/10

Fail

Unsupported Claims

- Cloud computing means storing files on Google Drive.

Response #7

Question

What is Python used for?

AI Response

Python is mainly used for web development and can also be used for data science, automation, AI, scripting, and many other applications.

Reference Answer

Python is a general-purpose programming language used across web, AI, automation, data science, and more.

Dimension	Score
Relevance	9/10
Accuracy	9/10
Hallucination	7/10
Completeness	9/10
Overall	8.5/10

Needs Improvement

Unsupported Claims

- Python is mainly used for web development

Response #8

Question

What is an API?

AI Response

API stands for Application Programming Interface. It defines rules that allow different software applications to communicate with each other.

Reference Answer

An API enables communication between software systems through defined interfaces.

Dimension	Score
Relevance	9/10
Accuracy	9/10
Hallucination	7/10
Completeness	9/10
Overall	8.5/10

Needs Improvement

Unsupported Claims

- API defines rules that allow different software applications to communicate with each other.

Response #9

Question

What is deep learning?

AI Response

Deep learning is a type of machine learning that uses multi-layer neural networks to learn complex patterns from large datasets.

Reference Answer

Deep learning uses neural networks with multiple layers to model complex relationships.

Dimension	Score
Relevance	9/10
Accuracy	9/10
Hallucination	7/10
Completeness	9/10
Overall	8.5/10

Needs Improvement

Unsupported Claims

- Deep learning is a type of machine learning that uses multi-layer neural networks to learn complex patterns from large datasets.

Response #10

Question

What is Git?

AI Response

Git is a social media platform for programmers where people chat and share photos.

Reference Answer

Git is a distributed version control system used to track changes in source code and collaborate.

Dimension	Score
Relevance	1/10
Accuracy	0/10
Hallucination	7/10
Completeness	1/10
Overall	2.2/10

Fail

Unsupported Claims

- Git is a social media platform for programmers where people chat and share photos.

Flagged Responses

Response #2

Verdict: Needs Improvement

Overall Score: 8.2/10

Unsupported Claims

- Machine learning is a branch of AI where systems learn patterns from data instead of relying only on explicitly programmed rules.

Response #3

Verdict: Needs Improvement

Overall Score: 7.6/10

Unsupported Claims

- Overfitting happens when a model memorizes the training data, including noise, and performs poorly on unseen data.

Response #4

Verdict: Needs Improvement

Overall Score: 8.2/10

Unsupported Claims

- Normalization scales numerical features to a common range, helping many algorithms converge faster and perform better.

Response #5

Verdict: Needs Improvement

Overall Score: 7.6/10

Unsupported Claims

- A database index is like a table of contents that speeds up data retrieval, although it may slightly increase storage and update costs.

Response #6

Verdict: Fail

Overall Score: 3.4/10

Unsupported Claims

- Cloud computing means storing files on Google Drive.

Response #7

Verdict: Needs Improvement

Overall Score: 8.5/10

Unsupported Claims

- Python is mainly used for web development

Response #8

Verdict: Needs Improvement

Overall Score: 8.5/10

Unsupported Claims

- API defines rules that allow different software applications to communicate with each other.

Response #9

Verdict: Needs Improvement

Overall Score: 8.5/10

Unsupported Claims

- Deep learning is a type of machine learning that uses multi-layer neural networks to learn complex patterns from large datasets.

Response #10

Verdict: Fail

Overall Score: 2.2/10

Unsupported Claims

- Git is a social media platform for programmers where people chat and share photos.

Improvement Recommendations

- Improve factual correctness and verify important claims before generating responses.
- Hallucination frequency is relatively high. Increase retrieval grounding and reference verification.

End of Report

Generated by
AI Response Quality Evaluator